

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I **M. Meghana (192525435)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **M. Meghana**

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly	Uses relevant	Uses vague or	No useful

		relevant cloud examples to strengthen the answer	examples with minor mismatch	partially relevant examples	examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Q1: Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Definition: cloud computing is the delivery of computing services such as servers , storage, database , networking , software ,and analytics over the internet.It allows users to access resource on demand without owning physical infrastructure

1. On-Demand Self-Service

- Users can access computing resources whenever needed without human intervention.
- Example: Creating a virtual machine on AWS.

2. Broad Network Access

- Services are available over the internet and can be accessed using laptops, smartphones, and tablets.
- Supports access from anywhere.

3. Resource Pooling

- Cloud providers share computing resources among multiple users using a multi-tenant model.
- Resources are allocated dynamically.

4. Rapid Elasticity

- Resources can be increased or decreased quickly based on demand.
- Example: An e-commerce website increases server capacity during sales.

Advantages

- Cost-effective
- Scalable
- Flexible
- High availability

Conclusion:

Cloud computing offers flexible, scalable, and cost-efficient services, making it more efficient than traditional on-premise computing.

Q2 .Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Evolution of Cloud Computing:

Cloud computing evolved through advances in hardware, networking, distributed systems, and internet technologies.

1. Hardware Evolution

- Large mainframe computers were used in the early days.
- Later, personal computers became popular.

2. Internet Evolution

- High-speed internet enabled global connectivity.
- Users could access remote systems easily.

3. Distributed Computing

- Multiple computers worked together to complete tasks.
- Improved performance and reliability.

4. Virtualization Technology

- Multiple virtual machines could run on one physical server.
- Improved resource utilization.

5. Web Services

- SOAP and REST enabled communication between applications over the internet.

6. Modern Cloud Platforms

- Companies like AWS, Microsoft Azure, and Google Cloud provide scalable cloud services.

Conclusion

The evolution from hardware systems to internet technologies and virtualization laid the foundation for today's cloud computing platforms.

Q3 .Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Defiination: Virtualization is the technology of creating virtual versions of physical resources such as servers, storage, networks, and desktops. It allows multiple virtual machines to run on one physical server

Types of virtualization

1.Server Virtualization:

A single physical server is divided into multiple virtual servers.

- Better hardware utilization
- Lower cost

2. Storage Virtualization

Combines storage from multiple devices into one logical storage unit.

- Easier management
- Better performance

3.Network Virtualization:

Creates virtual networks independent of physical hardware.

- Improved flexibility
- Better network management

4.Desktop Virtualization :

Users can access their desktop remotely from any device.

- Supports remote work
- Easy maintenance

Advanatages:

- Reduces hardware cost
- Saves energy
- Improves resource utilization
- Easy backup and recovery
- Supports cloud computing

Disadvantages

- Initial setup cost
- Requires skilled administrators
- Performance depends on hardware

Virtualization	Cloud Computing
Technology	Service model
Creates virtual machines	Delivers services over the internet
Uses Hypervisor	Uses virtualization plus management software
Focuses on hardware efficiency	Focuses on service delivery
Example: VMware	Example: AWS EC2

Virtualization is the backbone of cloud computing because it improves efficiency, reduces costs, and enables flexible resource management.

Q4.Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.

Feature	SOAP	REST
Full Form	Simple Object Access Protocol	Representational State Transfer
Communication	Protocol	Architectural style
Data Format	XML only	JSON, XML, HTML
Speed	Slower	Faster
Security	High (WS-Security)	HTTPS based
Complexity	More complex	Simple
Performance	Lower	Higher
Cloud Suitability	Enterprise applications	Cloud and mobile applications

Advantages of SOAP:

- High security
- Reliable messaging
- Suitable for banking and enterprise systems

Advantages of REST:

- Lightweight
- Fast
- Easy to develop
- Best for cloud applications and mobile apps

Conclusion

REST is widely used in modern cloud computing because it is lightweight, faster, and supports multiple data formats

Q5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Service Model	Meaning	Provider Gives	User Manages	Example
IaaS	Infrastructure as a Service	Virtual machines, storage, networking	OS, applications, data	AWS EC2
PaaS	Platform as a Service	Runtime, database, development tools	Application and data	Google App Engine
SaaS	Software as a Service	Complete software over internet	Only uses software	Gmail
XaaS	Anything as a Service	Any IT service through cloud	Depends on service	Microsoft 365

IaaS (Infrastructure as a Service)

- Provides virtual servers, storage, and networking.
- Users install and manage the operating system and applications.
- **Example:** AWS EC2.

PaaS (Platform as a Service)

- Provides a platform for developing and deploying applications.
- Developers focus only on coding.
- **Example:** Google App Engine.

SaaS (Software as a Service)

- Ready-to-use software accessed through a web browser.
- No installation or maintenance required.
- **Example:** Gmail.

XaaS (Anything as a Service)

- Refers to any cloud-delivered service such as Storage as a Service, Database as a Service, or Security as a Service.
- **Example:** Microsoft 365.

IaaS provides infrastructure, PaaS provides a development platform, SaaS provides ready-to-use software, and XaaS represents any service delivered through the cloud. These models offer flexibility, scalability, and cost savings for different user needs.